Public Research Preprints

Russell, D. R. (2026). **Molecular graph models for ADMET: multitask comparisons, ensemble probabilities, and retrospective sequential testing.** Public preprint, SNPTX.

Manuscript:
https://github.com/snptx1/snptx-repro-discovery/blob/main/preprint/dl_forward/MANUSCRIPT_learned_representations_admet.md

Reproducible walkthrough:
https://snptx1.github.io/snptx-academic/pages/notebooks/walkthrough_learned_representations_admet.html

Russell, D. R. (2026). **Knowing what to measure and when to stop: an autonomous decision engine for molecular property discovery.** Public preprint, SNPTX.

Manuscript:
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Research Website and Reproducibility Resources

SNPTX: An Experimentation Layer for Multi-Modal Biomedical Machine Learning. Research website and technical documentation covering system architecture, methods, results, limitations, and reproducible walkthroughs.

Research website:
https://snptx1.github.io/snptx-academic/

Companion code and reproducibility artifacts:
https://github.com/snptx1/snptx-repro-discovery

Planned Graduate Research

Independent Confirmation in Adaptive Molecular Discovery

Planned Harvard ALM capstone research, Spring 2027
Advisor: Stephen Elston, PhD

Building on SNPTX, this proposed project examines how adaptive molecular-discovery controllers should allocate limited measurements between exploration and independent confirmation. Using sequentially revealed outcomes from public assay datasets, it would compare evidence-reuse, reserved-confirmation and applicable selection-aware protocols, measuring incorrect decisions, missed discoveries and total measurement cost. Intended outputs are a reproducible benchmark and a manuscript suitable for submission.

The project would seek collaboration with an academic research laboratory within the Harvard ecosystem, subject to partner agreement and capstone approval.

Molecular graph models for ADMET: multitask comparisons, ensemble probabilities, and retrospective sequential testing

Daniel R. Russell | SNPTX | 2026 | Public preprint

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Abstract

A companion study built a sequential decision engine around a random-forest oracle. This paper asks what learned graph models add when they take the oracle's place, and answers by audit as much as by benchmark. On six Therapeutics Data Commons ADMET endpoints, under within-endpoint Murcko scaffold splits repeated over five seeds and four retained training-scaffold fractions, a multi-task Graph Isomorphism Network lowers Caco2 MAE by 0.11 to 0.17 relative to its single-task counterpart (18 of 20 seed-fraction pairs) and lifts HIA AUROC by 0.071 at fraction 0.50 (5 of 5 seeds); at full data it loses ground on AMES and Solubility (AUROC −0.049, MAE +0.145). We read these as comparisons of training protocols rather than measurements of transfer, because multi-task training recycles the smaller task loaders and, since splits are drawn per endpoint, the shared trunk trains on structures identical to 10-23% of each target's test records at full data. Turning from accuracy to uncertainty, a five-member temperature-scaled ensemble has the lowest graph-model NLL on one fixed classification split (0.482 against 0.505 for a single network and 0.532 for Monte Carlo dropout) and the lowest AURC, but a higher ECE; the descriptor random forest leads on accuracy and on most uncertainty metrics. The ensemble's nominal 90% prediction sets cover 0.877 of test labels on average, a figure we report as empirical rather than guaranteed, since temperature fitting reuses the conformal calibration labels and group splitting does not establish molecule-level exchangeability. Wired into the engine, a probability-ranked SPRT replay reaches GO on BBB and AMES after 105 measurements against an analytic fixed-sample budget of about 137.4; it runs under a Gaussian working model, standardizes with test-pool statistics, and sees one measurement order, so what it demonstrates is traceability, not prospective error control. Finally, an audit of a GAT attribution probe finds its atom score equal to one by softmax normalization, and we withdraw the salience interpretation. An executable walkthrough reconstructs every reported table from committed artifacts; retraining is a separate GPU tier.

Knowing what to measure and when to stop: an autonomous decision engine for molecular property discovery

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Abstract

Molecular property discovery is bottlenecked by the cost of measurement and by a trust gap: models rarely say how confident they are, or when enough has been measured to make a call. We present a calibrated, autonomous, sequential decision engine that turns ADMET property discovery into three coupled decisions and wires them into one closed loop. First, it reaches go/no-go calls at matched nominal error rates with materially fewer measurements using Wald's sequential probability ratio test (SPRT): at the empirical operating point (a 0.25 sigma shift) it needs 65 versus 99 measurements (a 34% saving) and saves 23% on average across effect sizes, with mean realized power 0.85 and Type I error 0.043 across the sweep. Second, it quantifies what it does not know with split-conformal prediction evaluated under leakage-controlled (Murcko scaffold) shift: empirical coverage is near, but for two of three endpoints below, the nominal 0.90 target (BBB 0.884, hERG 0.869, AMES 0.899); BBB random-split ECE is 0.044. Third, it abstains wisely: selective prediction lifts retained accuracy from 0.864 to 0.94 at 70% coverage under both random and scaffold splits. Wired into a closed loop with a pluggable oracle, a novelty archive, an abductive discovery cycle, and DuckDB provenance, the engine runs an end-to-end autonomous campaign that reaches six traceable go/no-go decisions using 140 versus 357 fixed-sample measurements (61% fewer); a companion discovery probe recovers an established structure-property driver on five of five probed endpoints (four independent drivers plus one closely related descriptor-endpoint reference) and surfaces 1398 interpretable structure-property cliffs on its two regression endpoints. All results are reproducible from a tagged commit on CPU.